DEEP VEIN THROMBOSIS (DVT)

THROMBUS FORMATION WITHIN THE DEEP VENOUS SYSTEM

WHAT DVT IS

DVT is a blood clot that forms in a deep vein—most commonly in the leg's deep venous system.

It arises from Virchow's triad: venous stasis, endothelial injury, and hypercoagulability.

Proximal (popliteal or above) clots carry the highest risk of pulmonary embolism (PE) and are part of the spectrum of venous thromboembolism (VTE).

WHY IT MATTERS

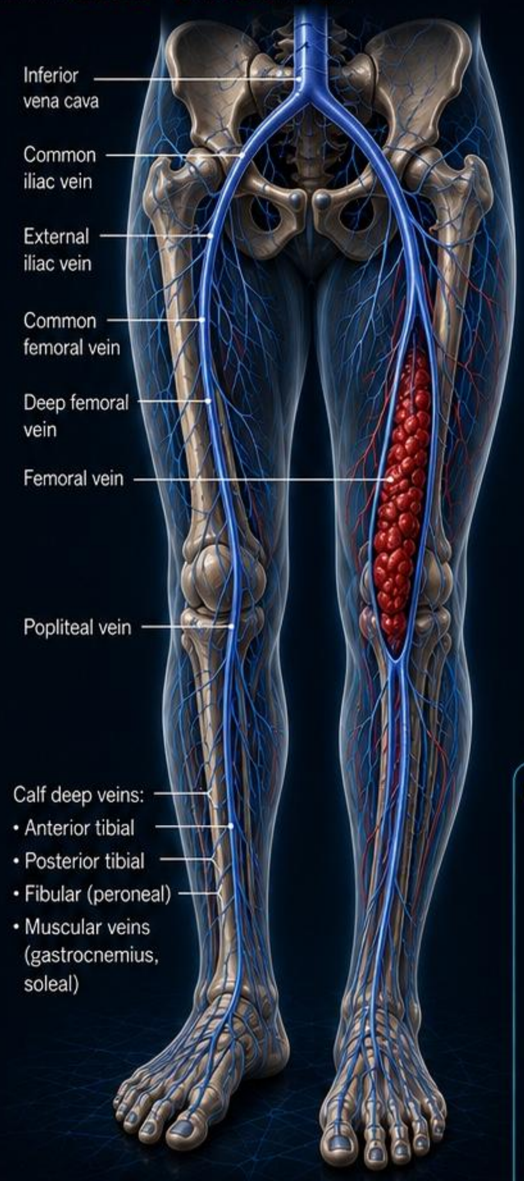
DVT can cause pain, swelling, and limb complications and may break free and travel to the lungs, causing a life-threatening PE.

Early recognition, accurate diagnosis, and prompt anticoagulation save lives and prevent long-term complications.

HOW A VENOUS THROMBUS FORMS

Low blood flow in a vein—often at a valve pocket—leads to blood stasis. This allows clotting factors and blood cells to accumulate and form a thrombus that can grow and propagate.

Thrombus typically begins in a valve pocket or low-flow segment of a deep vein.



HEALTHY FLOW
Normal venous return

DVT: REDUCED FLOW
Obstructed by thrombus

① VENOUS STASIS

Immobility, surgery, travel, or other factors slow venous return, allowing blood to pool and clot.

② THROMBUS PROPAGATION

The clot can extend proximally within the deep venous system, increasing the risk of complications.

③ PULMONARY EMBOLIZATION RISK

A fragment of the thrombus can detach, travel through the venous system, and lodge in the pulmonary arteries—causing a potentially fatal pulmonary embolism.

AT A GLANCE

- Below the popliteal vein
- Lower risk of PE
- May be managed with surveillance or anticoagulation based on risk and symptoms

PROXIMAL (POPLITEAL AND ABOVE) DVT

- Popliteal vein and above
- Highest risk of PE
- Anticoagulation recommended
- Greater risk of long-term complications (post-thrombotic syndrome)

DVT most commonly affects the lower extremities but can occur in the pelvis or upper extremities.

EDUCATIONAL PEARL

"A noncompressible deep vein confirms the clot; its location determines much of the risk."

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Overview of Deep Vein Thrombosis	Clinical Overview	3D Medical Illustration	2021 CHEST Guideline – Antithrombotic Therapy for VTE Disease (2nd Update)	CLINICAL PENDING

LOWER-EXTREMITY VENOUS ANATOMY

DEEP VENOUS SYSTEM OF THE LOWER EXTREMITY

DEEP VEINS

Common iliac vein
Formed by the union of the internal and external iliac veins; drains the pelvis.

External iliac vein
Continuation of the femoral vein above the inguinal ligament.

Common femoral vein
Formed by the union of the great saphenous vein and deep femoral vein.

Deep femoral vein (Profunda femoris)
Drains the deep muscles of the thigh.

Femoral vein
The main deep vein of the thigh; continues through the adductor canal.

Popliteal vein
Located in the popliteal fossa behind the knee.

Tibial veins
Anterior tibial, posterior tibial, and peroneal veins accompany the corresponding arteries.

Gastrocnemius veins
Drain the gastrocnemius muscle; important sources of calf DVT.

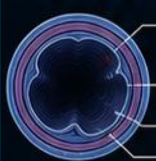
VENOUS TERRITORIES

- Iliac veins
- Femoral veins
- Popliteal vein
- Tibial veins
- Foot veins

VALVULAR STRUCTURE

Veins contain valves that ensure unidirectional blood flow toward the heart.

CROSS-SECTION (TYPICAL VEIN)



AT A GLANCE

The deep venous system begins in the foot veins and ascends through the tibial, popliteal, femoral, and iliac veins to the inferior vena cava. Valves and the muscle pump of the calf work together to return blood to the heart.

DEEP SYSTEM
Deep veins accompany arteries and are protected by surrounding muscle.

VALVES
Valves prevent backflow and maintain unidirectional flow.

CLINICAL RELEVANCE
Common sites of DVT include the calf veins, popliteal vein, and femoral vein.

EDUCATIONAL PEARL

"Understanding the deep venous anatomy is essential—DVT can occur anywhere in this system, from the calf veins to the iliac veins."

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Lower-Extremity Venous Anatomy	Clinical Overview	3D Medical Illustration	2021 CHEST Guideline – Antithrombotic Therapy for VTE Disease (2nd Update)	CLINICAL PENDING

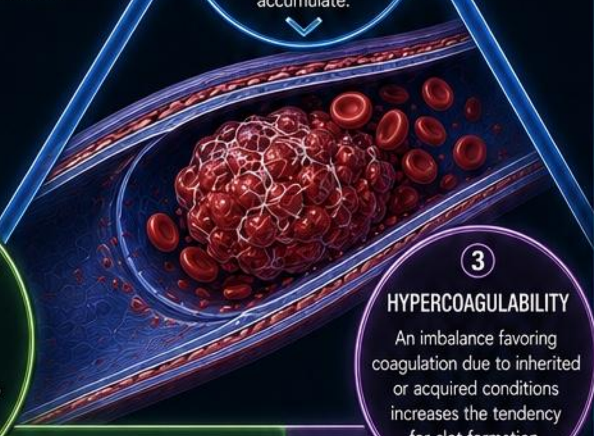
PATHOPHYSIOLOGY OF DVT

VIRCHOW'S TRIAD: THE THREE PILLARS OF THROMBUS FORMATION

Deep vein thrombosis (DVT) occurs when the balance between procoagulant and anticoagulant forces is disrupted within the deep venous system.

Virchow's triad identifies the three primary factors that promote thrombus formation.

1 VENOUS STASIS
Reduced blood flow causes pooling of blood allowing clotting factors and platelets to accumulate.



2 ENDOTHELIAL INJURY OR ACTIVATION
Endothelial injury or activation promotes tissue factor expression, inflammation, and coagulation.

3 HYPERCOAGULABILITY
An imbalance favoring coagulation due to inherited or acquired conditions increases the tendency for clot formation.

COMMON RISK FACTORS

- Surgery or trauma
- Immobility or prolonged bed rest
- Long-distance travel
- Malignancy
- Pregnancy / postpartum
- Hormone therapy (eg, OCP, HRT)
- Obesity
- Smoking
- Inherited thrombophilia (eg, Factor V Leiden, Prothrombin mutation)
- Prior DVT or PE

Risks often act together. The more factors present, the higher the likelihood of thrombosis.

THROMBUS FORMATION CASCADE (TYPICAL IN DVT)

1. INITIATING FACTORS
Stasis, endothelial injury/activation, or hypercoagulability trigger coagulation.

2. THROMBIN GENERATION
Activation of the coagulation cascade leads to thrombin production.

3. FIBRIN NETWORK FORMATION
Thrombin converts fibrinogen to fibrin, forming a structural scaffold.

4. CELLULAR TRAPPING
Red blood cells and leukocytes are trapped within the fibrin network.

5. THROMBUS PROPAGATION
The clot extends proximally and may increase in size.

Venous thrombi are rich in fibrin and red blood cells with relatively fewer platelets.

KEY POINT

DVT results from an interaction of venous stasis, endothelial injury/activation, and hypercoagulability.

Virchow's triad explains why risks often compound; reducing modifiable components lowers the likelihood of thrombosis and recurrence.

HEMOSTATIC BALANCE

NORMAL (BALANCED HEMOSTASIS)

DVT (HEMOSTATIC IMBALANCE)

Net shift toward coagulation promotes clot formation and propagation.

- Virchow's triad explains the major mechanisms behind DVT.
- Disruption of the hemostatic balance leads to thrombus formation.
- Multiple risk factors increase the likelihood of thrombosis and recurrence.
- Understanding pathophysiology guides prevention and treatment.

EDUCATIONAL PEARL

"DVT is the end result of Virchow's triad—addressing any one component can tip the balance back toward normal."

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Pathophysiology of DVT	Pathophysiology Overview	3D Medical Illustration	2021 CHEST Guideline – Antithrombotic Therapy for VTE Disease (2nd Update)	CLINICAL PENDING

CLINICAL PRESENTATION OF DVT

RECOGNIZING THE SIGNS AND SYMPTOMS

DVT may range from asymptomatic to severe. Symptoms typically develop over hours to days and usually affect one leg.

KEY CLINICAL FEATURES

LEG SWELLING
Usually unilateral; may involve the entire leg.

PAIN OR TENDERNESS
Often in the calf or thigh; may feel like a cramp or soreness.

WARMTH
Affected leg may feel warmer than the opposite side.

REDNESS OR DISCOLORATION
May be subtle; due to enhanced venous pressure and inflammation.

VISIBLE SUPERFICIAL VEINS
Collateral veins may become more prominent.

HOMAN SIGN* (HISTORICAL)
Calf pain with passive dorsiflexion of the foot. Low sensitivity and specificity; not recommended alone for diagnosis.

*Not recommended as a standalone diagnostic test.

CONSIDER OTHER CAUSES

Muscle strain
(Localized pain, no swelling)

Baker cyst
(Popliteal fullness or pain)

Cellulitis
(Usually with fever, erythema, warmth & tenderness)

Chronic venous insufficiency
(Bilateral leg swelling, hyperpigmentation)

Lymphedema
(Non-pitting swelling, chronic course)

RED FLAGS – THINK PE

- Sudden shortness of breath
- Chest pain (pleuritic)
- Cough, possibly with blood (hemoptysis)
- Rapid heart rate
- Lightheadedness or syncope

Seek immediate medical attention.

UP TO 50% OF DVT CASES MAY BE ASYMPTOMATIC.
High clinical suspicion is critical in at-risk patients.

HIGHER CLINICAL SUSPICION IN PATIENTS WITH RISK FACTORS

- Recent surgery (especially orthopedic)
- Active cancer
- Prolonged immobility
- Previous VTE
- Pregnancy or postpartum
- Hormone therapy (eg, OCP, HRT)
- Thrombophilia
- Increasing age

AT A GLANCE

DVT classically presents with unilateral leg swelling, pain, warmth, and tenderness. However, presentation can be subtle or asymptomatic. Always assess risk factors and consider alternative diagnoses.

CLINICAL PEARL

When in doubt, rule it out.
Early recognition of DVT prevents complications such as pulmonary embolism and post-thrombotic syndrome.

DVT MOST COMMONLY OCCURS IN:

EDUCATIONAL PEARL

"Clinical assessment is the first step—use validated tools and imaging to confirm the diagnosis."

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Clinical Presentation of DVT	Pathophysiology Overview	3D Medical Illustration	2021 CHEST Guideline – Antithrombotic Therapy for VTE Disease (2nd Update)	CLINICAL PENDING

PHYSICAL EXAMINATION FOR DVT

A SYSTEMATIC APPROACH TO THE AFFECTED LEG

No single physical finding can rule in or rule out DVT. Clinical assessment should be combined with pretest probability tools and diagnostic imaging.

KEY EXAM COMPONENTS

INSPECTION
Look for swelling, erythema, discoloration, collateral veins, and visible superficial veins.

PALPATION
Palpate gently along the deep venous distribution (posterior calf, along femoral vein course) for tenderness or cord-like structures.

MEASUREMENT
Compare calf circumference (10 cm below tibial tuberosity) bilaterally.

DISTAL NEUROVASCULAR
Check distal pulses, motor function, sensation, and capillary refill.

WARMTH
Affected leg often feels warmer than the opposite side.

SUPERFICIAL VEINS
Dilated superficial veins may be present.

Affected leg

COMMON FINDINGS

CALF SWELLING
Usually unilateral (>3 cm larger than opposite leg).

WARMTH
Affected leg feels warmer to touch.

TENDERNESS
Most common along deep veins of the calf or thigh.

DILATED SUPERFICIAL VEINS
Collateral venous circulation may be visible.

ERYTHEMA
May be mild and non-specific.

FINDINGS THAT ARE NOT SPECIFIC

- Pitting edema (non-specific)
- Homan sign (low sensitivity & specificity)
- Mild discomfort (common with many benign conditions)

These findings alone do not confirm DVT.

WELLS CRITERIA FOR DVT

Criterion	Points
Active cancer (treatment within 6 months or palliative)	+1
Paralysis, paresis, or recent plaster immobilization	+1
Recently bedridden >3 days or major surgery within 12 weeks	+1
Localized tenderness along the deep venous system	+1
Entire leg swollen	+1
Calf swelling ≥3 cm compared to asymptomatic leg (10 cm below tibial tuberosity)	+1
Pitting edema limited to symptomatic leg	+1
Collateral superficial veins (non-varicose)	+1
Previously documented DVT	+1
Alternative diagnosis at least as likely as DVT	−2

Validated Wells score for DVT.
Reference: Wells PS, et al. Ann Intern Med. 2003;138:98-107.

SCORE	INTERPRETATION
≤1	DVT UNLIKELY
≥2	DVT LIKELY

Use pretest probability (e.g., Wells score) to guide need for D-dimer testing and imaging.

EXAM PEARL

A normal physical exam does not exclude DVT. Maintain a high index of suspicion in at-risk patients, especially with unilateral calf or thigh swelling and tenderness.

AT A GLANCE

- Compare both legs systematically. Inspect, palpate, measure.
- Combine exam findings with pretest probability tools.
- Use results to determine need for D-dimer testing and imaging.
- Early and accurate assessment improves outcomes and prevents complications.

EDUCATIONAL PEARL

"The physical exam provides clues, not answers—use it to estimate probability and guide the next best test."

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Physical Examination for DVT	Clinical Assessment	Clinical Photography	ASH VTE Guidelines Diagnostic Testing for DVT (2018)	CLINICAL PENDING

DIAGNOSTIC IMAGING FOR DVT

ACCURATE, TIMELY IMAGING GUIDES TREATMENT AND PREVENTS COMPLICATIONS.

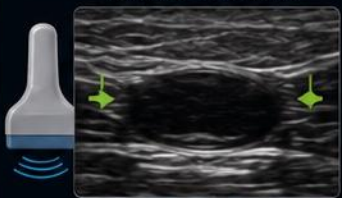
Compression ultrasonography (CUS) is the first-line test for suspected DVT.
It is rapid, noninvasive, bedside-friendly, and highly accurate for proximal DVT.

FIRST-LINE TEST: COMPRESSION ULTRASONOGRAPHY (CUS)

High sensitivity and specificity for proximal (above-knee) DVT.

NORMAL VEIN (COMPRESSIBLE)

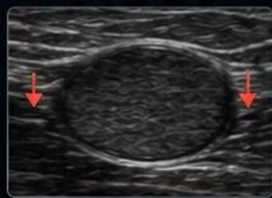
Complete compression obliterates the lumen—no thrombus.



Vein collapses with compression

DVT – NONCOMPRESSIBLE

Thrombus prevents compression; lumen remains open.

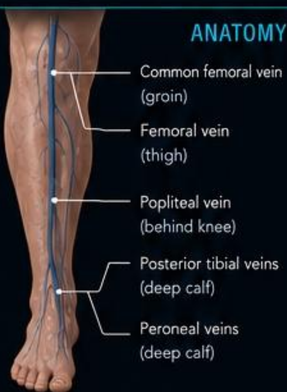


Vein remains open with compression

CUS PERFORMANCE (PROXIMAL DVT)

- Sensitivity ~97%
- Specificity ~94%
- Operator dependent and limited for distal (calf) DVT

ANATOMY ASSESSED BY CUS



COMMON FEMORAL VEIN
Assess compressibility just below the inguinal ligament.

FEMORAL VEIN
Scan along the entire thigh to the adductor canal.

POPLITEAL VEIN
Assess in the popliteal fossa from multiple views.

CALF VEINS
Examine posterior tibial and peroneal veins. Calf DVT is less predictable and more likely to require follow-up.

IMAGING MODALITIES AT A GLANCE

MODALITY	BEST FOR	ADVANTAGES	LIMITATIONS
Compression Ultrasonography (CUS)	First-line evaluation of suspected DVT	• No radiation • Bedside available • High accuracy for proximal DVT • Rapid and inexpensive	• Operator dependent • Limited for pelvic/iliocaval DVT • Limited for isolated calf DVT
CT Venography (CTV)	Pelvic/iliac DVT or when CUS is inconclusive	• Excellent anatomic detail • Rapid acquisition	• Radiation exposure • Iodinated contrast (nephrotoxicity, allergy risk)
MR Venography (MRV)	Contrast allergy or renal insufficiency	• No ionizing radiation • Good for pelvic/iliac veins	• Less available, longer scan time • Higher cost
Conventional Venography	When intervention is planned or diagnosis remains uncertain	• Highest spatial resolution • Allows immediate intervention	• Invasive • Contrast risk
D-dimer Blood Test	Exclude DVT in low-risk (Wells ≤1) outpatients	• Highly sensitive • Widely available, low cost	• Not specific • Many false positives

WHEN CUS MAY BE INCONCLUSIVE

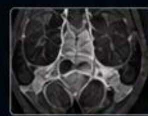
- Poor acoustic windows (obesity, edema, dressings)
- Isolated distal (calf) vein thrombosis
- Inability to visualize entire deep venous system
- Discrepant clinical probability and CUS results

RECOMMENDATION

If initial CUS is negative but clinical suspicion remains moderate or high, repeat CUS in 5–7 days to detect propagation of a distal clot.

ADJUNCT IMAGING (WHEN NEEDED)

CT VENOGRAPHY (CTV)
Excellent for pelvic or iliocaval thrombosis. Useful when ultrasound is inconclusive or when proximal extension is suspected.



MR VENOGRAPHY (MRV)
No ionizing radiation. Useful for patients with iodinated contrast allergy or renal insufficiency.

CONVENTIONAL VENOGRAPHY
Historically the gold standard. Now reserved for selected cases when endovascular intervention is planned.

D-DIMER BLOOD TEST
Highly sensitive but not specific. Best used to exclude DVT in patients with LOW pretest probability. Not useful in moderate/high probability or inpatients.

KEY POINT

- Compression ultrasonography is the cornerstone for diagnosing DVT.
- Use the right test at the right time based on clinical probability.
- Repeat imaging is essential when initial studies are negative but clinical suspicion persists.
- Accurate diagnosis prevents unnecessary treatment and reduces the risk of complications.

PRETEST PROBABILITY GUIDES TESTING

LOW (Wells ≤1) → D-DIMER → MODERATE/HIGH (Wells ≥2)

D-DIMER: If negative, DVT excluded

MODERATE/HIGH: Proceed directly to imaging. D-dimer is not helpful when clinical probability is moderate or high.

EDUCATIONAL PEARL

"Use the right test at the right time—integrate clinical probability, ultrasound findings, and follow-up imaging when needed."

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Diagnostic Imaging for DVT	Diagnostic Evaluation	Ultrasound, CT, MR, Lab	ASH VTE Guidelines – Antithrombotic Therapy for VTE Disease (2nd Update), 2021; ACCP Guidelines, Chest 2021 (CUS)	CLINICAL PENDING

GROSS PATHOLOGY OF DVT

THROMBUS CHARACTERISTICS AND ANATOMIC DISTRIBUTION

DVT thrombi vary in age, consistency, and appearance. Understanding their gross features helps explain complications and guides management.

TYPICAL APEARANCE BY MATURATION STAGE

ACUTE (EARLY)

- Dark red, soft, gelatinous
- Formed from fresh clot
- Poorly adherent to vessel wall

ORGANIZING (SUBACUTE)

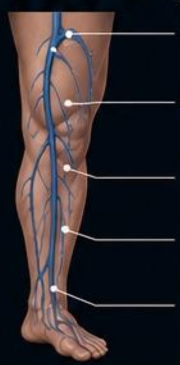
- Red to brown
- Firmer, more organized
- Adherent to vessel wall
- Recanalization may begin

CHRONIC / ORGANIZED

- Tan to white, fibrous
- Firm, organized
- Recanalized channels may be present

ANATOMIC DISTRIBUTION OF DVT

DVT most commonly involves the deep veins of the lower extremity.



COMMON SITES OF INVOLVEMENT*

- Iliac veins (pelvic)
- Femoral vein (thigh)
- Popliteal vein (behind knee)
- Tibial veins (calf)
- Peroneal veins (calf)

*Distribution varies by population and study.

THROMBUS FEATURES

Location: Most often in the proximal veins (popliteal, femoral, iliac). Proximal DVT carries higher risk of PE.

Extension: May extend longitudinally and can be occlusive or nonocclusive.

Adherence: Organizing thrombi become increasingly adherent to the venous wall.

Recanalization: Fibrovascular channels may restore partial venous flow through organized thrombus.

Embolization risk: Portions of an acute or subacute thrombus can embolize to the pulmonary arteries.

SUPERFICIAL VENOUS THROMBOSIS

Superficial venous thrombi are palpable, tender, and located in superficial veins (e.g., great saphenous vein). They have a lower risk of PE but may extend toward the deep system.

POTENTIAL COMPLICATIONS OF DVT

LEG VEIN

Thrombus forms in deep veins of the lower extremity.

→ **INFERIOR VENA CAVA (IVC)**

Emboli can travel through the IVC to the right heart.

→ **RIGHT HEART**

Emboli may lodge in the right ventricle and pulmonary outflow tract.

→ **PULMONARY ARTERIES**

Emboli can obstruct pulmonary arteries and cause PE.

CLINICAL CORRELATION

- Proximal DVTs are more likely to cause PE and post-thrombotic syndrome.
- Post-thrombotic syndrome results from persistent venous obstruction and venous valve injury.
- Early diagnosis and anticoagulation reduce complications.
- Assess extent and location of thrombus on imaging to guide management and follow-up.

EDUCATIONAL PEARL

"The age and location of a thrombus influence symptoms, complications, and treatment duration."

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Gross Pathology of DVT	Pathology	Gross Specimen Photography, Illustration	ACCP Guidelines – Antithrombotic Therapy for VTE Disease (2nd Update), 2021; ASH VTE Guidelines, 2021	CLINICAL PENDING

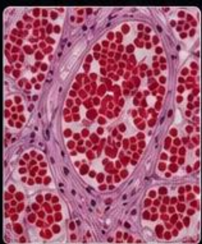
HISTOLOGY OF DVT

EVOLUTION OF THROMBUS THROUGH ORGANIZATION AND RECANALIZATION

The microscopic appearance of DVT evolves through organization and recanalization, although morphology cannot precisely determine clot age.

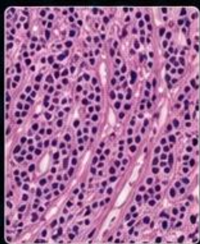
EVOLUTION OF DVT THROMBUS

ACUTE / EARLY



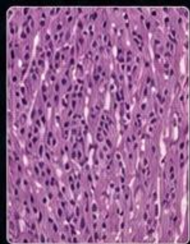
- Fibrin-rich network entraps erythrocytes, leukocytes, and a smaller platelet component
- Neutrophils present
- Minimal organization

ORGANIZING



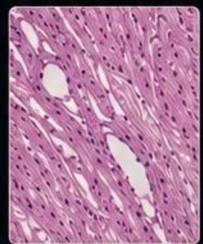
- Fibroblasts and macrophages infiltrate
- Fibrin is replaced by collagen
- Neovascularization begins

ORGANIZED



- Dense collagen and fibroblasts predominate
- Neovascularization increases
- Thrombus becomes firmer and adherent

CHRONIC / RECANALIZED

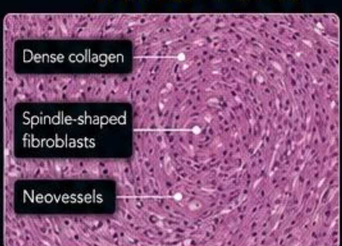


- Collagenous scar with recanalized channels
- Variable calcification may be present
- Minimal cellularity

PROGRESSIVE ORGANIZATION, ADHERENCE, AND RECANALIZATION

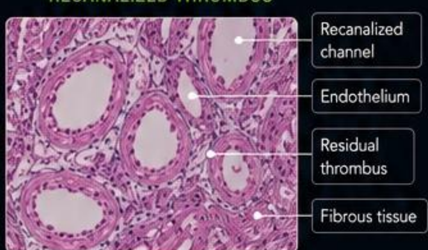
ORGANIZATION AND RECANALIZATION

ORGANIZED THROMBUS



Thrombus becomes incorporated into the venous wall and is firm and adherent.

RECANALIZED THROMBUS



Multiple endothelial-lined channels develop within the organized thrombus, restoring partial venous flow.

Histology is not routinely required for diagnosis but is valuable in research and when thrombus is removed for other reasons.

COMMON HISTOLOGIC STAINS

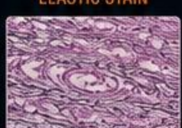
H&E (HEMATOXYLIN & EOSIN)

Excellent for overall architecture, cellularity, and inflammatory cells.

MASSON TRICHROME

Highlights collagen (blue) within organized thrombus.

ELASTIC STAIN



Demonstrates elastic fibers of the venous wall and helps evaluate chronic changes.

CD31 IHC

Highlights endothelial cells lining channels (recanalization) within the thrombus.

KEY HISTOLOGIC FEATURES

FIBRIN-RICH MATRIX
A fibrin network forms the scaffold of the thrombus and traps erythrocytes, leukocytes, and a smaller platelet component.

INFLAMMATORY CELLS
Neutrophils predominate early. Macrophages increase during organization and remove debris.

NEOVASCULARIZATION
New capillaries grow into the thrombus, supporting organization and the development of channels.

COLLAGEN DEPOSITION
Fibroblasts lay down collagen, replacing fibrin and forming scar-like tissue.

RECANALIZATION
Endothelial-lined channels develop within the organized thrombus, restoring partial venous flow.

CLINICAL CORRELATION

- Organized thrombi are more adherent and less likely to embolize than acute thrombi.
- Recanalization explains why some patients remain asymptomatic despite residual thrombus on imaging.
- Post-thrombotic syndrome results from chronic venous obstruction and valvular damage, not from the thrombus alone.
- Histologic changes parallel imaging findings and clinical presentation over time.

EDUCATIONAL PEARL

A venous thrombus evolves from a fibrin–erythrocyte clot into organized fibrous tissue; recanalization may restore partial flow but does not restore normal venous valves.

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Histology of DVT	Pathology	Histology (Stains, IHC)	Ageno W, et al. J Thromb Haemost. 2008;6(5):729-736. Undas A, et al. Thromb Haemost. 2010;103(1):89-96.	CLINICAL PENDING

TREATMENT OF DVT

GOALS: PREVENT THROMBUS EXTENSION, REDUCE RECURRENCE, AND PREVENT COMPLICATIONS

Management is guided by clinical scenario, bleeding risk, and patient-specific factors.

ANTICOAGULATION: CORNERSTONE OF THERAPY

AGENT	INITIATION	EXAMPLES	NOTES
DOACs (Preferred)	Start without parenteral anticoagulation	• Apixaban • Rivaroxaban	Preferred over warfarin for most patients without cancer-associated DVT or severe renal impairment.
DOACs (Requiring initial parenteral therapy)	Begin with parenteral anticoagulation for 5–10 days	• Dabigatran • Edoxaban	Then transition to full-dose DOAC.
Warfarin (Vitamin K antagonist)	Start with parenteral anticoagulation and overlap ≥5 days	• Warfarin	Continue overlap until INR is 2.0–3.0 for ≥24 hours.

DURATION OF THERAPY

- Minimum treatment phase: 3 months for all patients.
- Transiently provoked DVT (e.g., surgery, trauma, immobilization): stop after 3 months.
- Unprovoked DVT or persistent risk factors: evaluate for extended (potentially indefinite) anticoagulation if bleeding risk is not high.
- Reassess risks and benefits periodically.

ADJUNCTIVE MEASURES

Compression (for symptom relief)
Use graduated compression stockings (20–30 mmHg) for relief of leg swelling or discomfort. Routine use to prevent post-thrombotic syndrome is not recommended.

Early ambulation
Encourage early ambulation as tolerated and regular leg exercises.

Limb elevation
Elevate the affected leg when sitting or lying to reduce swelling.

Analgesia
Use acetaminophen or other analgesics as needed for pain control.

Adherence & precautions
Take anticoagulants as prescribed. Review bleeding precautions and avoid interacting medications.

Risk factor modification
Avoid smoking. Manage comorbidities (e.g., hypertension, diabetes, lipids). Maintain a healthy body weight.

INTERVENTIONAL OPTIONS (SELECTED PATIENTS)

CATHETER-DIRECTED THROMBOLYSIS (CDT)

For carefully selected patients with acute, symptomatic iliofemoral DVT (<14 days duration), low bleeding risk, and severe symptoms or limb threat. Not routine therapy.

MECHANICAL THROMBECTOMY

Mechanical removal of thrombus may be considered in the same selected patients as CDT. Not recommended for routine use in lower extremity DVT.

INFERIOR VENA CAVA (IVC) FILTER

Do not add an IVC filter to effective anticoagulation. Consider a retrievable filter for acute proximal DVT when anticoagulation is temporarily contraindicated. Remove as soon as feasible.

MANAGEMENT OF SPECIAL SITUATIONS

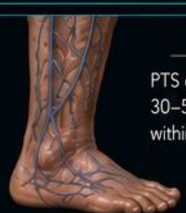
Pregnancy
- LMWH is the preferred treatment during pregnancy.
- Avoid DOACs and warfarin during pregnancy.
- Plan postpartum anticoagulation individually.
- LMWH is compatible with breastfeeding.

Cancer-associated DVT
- Use DOACs or LMWH for at least 3–6 months.
- Individualize based on bleeding risk, tumor type/location, drug interactions, renal function, and patient preference.

Renal impairment
- Adjust DOAC dose based on creatinine clearance.
- Consider LMWH or warfarin in severe renal dysfunction (varies by agent).

Elderly / High bleeding risk
- Individualize therapy.
- Balance risk of recurrent VTE against bleeding risk.
- Consider shorter duration if provoked and bleeding risk is high.

POST-THROMBOTIC SYNDROME (PTS): PREVENTION AND MANAGEMENT



PTS can occur in up to 30–50% of patients within 2 years after DVT.

RISK FACTORS
- Extensive proximal (iliofemoral) DVT
- Proximal location
- Recurrent ipsilateral DVT
- Persistent leg symptoms
- Inadequate anticoagulation
- Obesity
- Older age

PREVENTION
- Early and adequate anticoagulation
- Graduated compression stockings for symptom relief
- Weight management
- Regular exercise and ambulation
- Skin care and moisturization

MANAGEMENT
- Compression therapy
- Leg elevation
- Exercise and physical therapy
- Skin care
- Treat venous ulcers if present

KEY POINT

Prompt anticoagulation is the foundation of DVT treatment. Minimum treatment is 3 months, but duration should be individualized. Adjunctive measures and risk factor management support recovery and help reduce recurrence and complications.

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Treatment of DVT	Management	Algorithm / Illustration	ASH VTE Guidelines – Antithrombotic Therapy for VTE Disease (2nd Update), 2021; ACCP Guidelines, Chest, 2021	CLINICAL PENDING

CLINICAL PEARL: KEY TAKEAWAYS FROM DVT

RECOGNIZE EARLY. TREAT EFFECTIVELY. PREVENT RECURRENCE.

DVT is a common but potentially serious condition. Timely diagnosis, appropriate treatment, and attention to risk factors can prevent complications and improve outcomes.

1. RECOGNIZE THE KEY POINTS

- DVT most often occurs in the deep veins of the leg (calf, popliteal, femoral, and iliac).
- Classic symptoms include unilateral leg swelling, pain or tenderness, warmth, and erythema.
- Imaging with compression ultrasound is the first-line diagnostic test.
- Untreated DVT can lead to pulmonary embolism (PE) and post-thrombotic syndrome (PTS).

2. TREAT WITH EVIDENCE-BASED STRATEGIES

Anticoagulation is the foundation of treatment.
Choose the agent and duration based on clinical scenario and bleeding risk.

Most proximal DVTs—and distal DVTs selected for anticoagulation—require a primary treatment course of approximately 3 months.

Monitor, reassess, and adjust.
Reevaluate bleeding risk, patient preference, and adherence regularly.

Individualize care.
Special populations and comorbidities require tailored management.

3. PREVENT COMPLICATIONS

- Prevent PE with effective anticoagulation.
- Assess for and address bleeding risk.
- Consider compression and elevation for symptom relief.
- Be vigilant for and manage post-thrombotic syndrome (PTS).

KNOW THE RISK FACTORS

PATIENT FACTORS
- Increasing age
- Obesity
- Pregnancy / Postpartum
- Hormonal therapy (e.g., estrogen)
- Thrombophilia*
- Prior VTE
- Cancer

MEDICAL CONDITIONS
- Heart failure
- Inflammatory disease
- Nephrotic syndrome
- Central venous catheter
- Recent infection

SURGICAL / TRAUMA
- Major surgery (especially orthopedic)
- Trauma or fracture
- Immobility > 3 days

HOSPITAL / ACQUIRED
- Prolonged hospitalization
- ICU stay
- ICU lines

LIFESTYLE / OTHER
- Prolonged immobility (travel, sedentary)
- Smoking
- Dehydration* (possible contributor during illness / immobility)

*Thrombophilia testing is selective and should be ordered only when the result could change management. *Less well-established than other listed risk factors.

RED FLAGS: SEEK IMMEDIATE CARE

- Sudden shortness of breath
- Chest pain that worsens with deep breathing
- Coughing up blood
- Rapid heart rate
- Fainting or severe lightheadedness
- Worsening leg pain, swelling, or discoloration despite treatment

REMEMBER:

Early recognition, appropriate treatment, and ongoing prevention save lives.

PREVENTION: REDUCE RISK

- Stay active: move your legs frequently.
- Hydrate adequately.
- Maintain a healthy weight.
- Avoid smoking.
- Follow medical advice and take anticoagulants as prescribed.
- Use compression for symptom relief if recommended.
- Manage chronic conditions effectively.
- Attend follow-up and adhere to therapy.

FOLLOW-UP: ESSENTIAL CHECKLIST

- Confirm adherence to anticoagulant therapy.
- Assess for bleeding or adverse effects.
- Reassess risk factors and need for extended therapy.
- Evaluate for symptoms of post-thrombotic syndrome.
- Review for signs or symptoms of recurrent DVT or PE.
- Reinforce education on prevention and lifestyle measures.

CLINICAL PEARL | *Many DVT events are preventable, and timely diagnosis and treatment substantially reduce complications and recurrence.*

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Clinical Pearl: Key Takeaways from DVT	Key Takeaways / Summary	Illustration / Infographic	ASH VTE Guidelines – Antithrombotic Therapy for VTE Disease (2nd Update), 2021; ACCP Guidelines, Chest, 2021	CLINICAL PENDING